



Incredible Hulk Comet

A green coloured "Incredible Hulk" comet zoomed past the Earth last month. <https://digit.in/sep18-70-1>



Stone tools older than man

Before the homo sapiens, another race made stone tools and crossed oceans, maybe even without boats. <https://digit.in/sep18-70-2>

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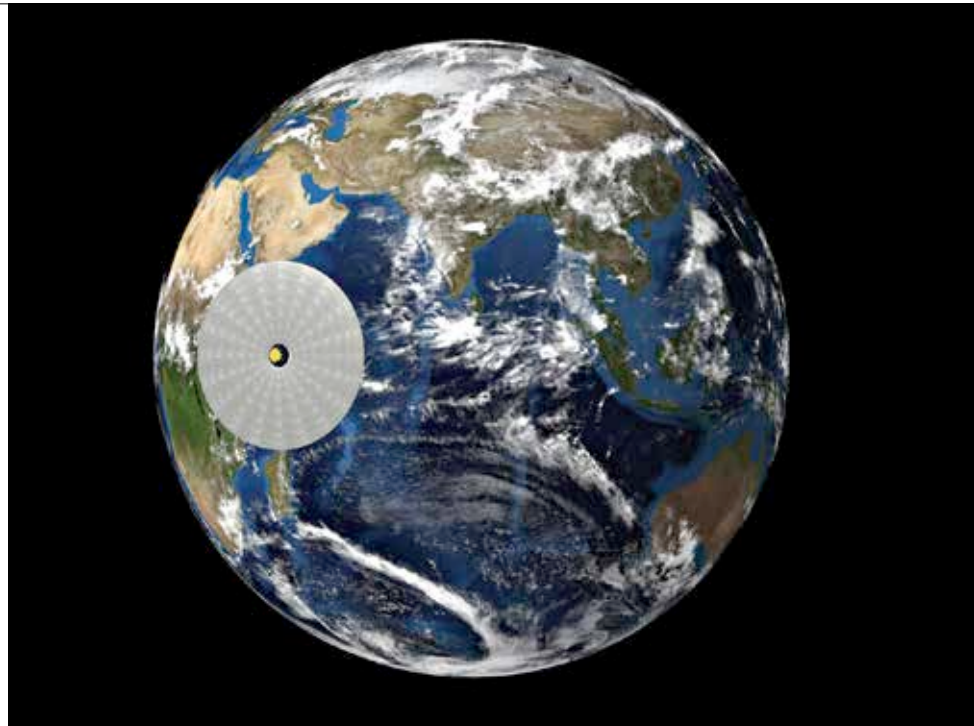
hen it comes to geoengineering the planet, the number of approaches can be summed up into two

broad categories. One approach is solar radiation management (SRM), which is using any technique to control the amount of heat received by the planet. The other is removing greenhouse gases, particularly carbon dioxide from the atmosphere. Both of these approaches are last ditch efforts, considered an "insurance policy", when reducing the amount of emissions does not do enough to check the rise in global temperatures. When all else fails, this is the Plan B.

SOLAR RADIATION MANAGEMENT

In 1991, the Mount Pinatubo volcano in the Philippines erupted and dumped about 17 million tons of sulphur dioxide in the atmosphere. It was the largest injection of aerosols in the atmosphere since the Krakatoa eruption in 1883. Because of the particles in the atmosphere, the amount of sunlight received by the Earth was reduced by 10 percent. As a direct result of this event, global temperatures reduced between 0.5 and 0.6°C. Additionally, in the following year, photosynthesis by plants was 23% higher than 1997, when all the dust had settled. This is because the fine particles in the atmosphere diffused the sunlight, allowing the light to reach the lower leaves of the trees and plants. As a consequence, the plants absorbed more carbon dioxide from the atmosphere, following the eruption. The Mount Pinatubo eruption demonstrated the capabilities of aerosols in the atmosphere to delay or reduce global warming.

Most of the techniques involve dumping chemicals into the atmosphere. Sulfur dioxide in large amounts mimics the effects of the Pinatubo eruption. Other candidates include aluminum oxide, common salt, calcium carbonate, moon dust and even diamond dust. The particles could be injected in the atmosphere through



Hacking

There are a number of approaches that scientists are considering to reverse global warming, but is geoengineering really a permanent solution to climate change?

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planes, drones, balloons, rockets, artillery, or pipes.

Then there are a number of proposed techniques to take the help of the clouds. Aerosols of sea salts injected into clouds would increase their water content, and make them more reflective. Introducing substances such as bismuth tri-iodide to the cirrus clouds in the upper atmosphere, would make the water to turn into ice, then reflect more of the sunlight back into space. Getting rid of the cirrus clouds is also a good idea, as they trap more heat than

they reflect. Spraying sea water on clouds would make them more reflective. Another approach, is to just create more clouds.

There are issues with dumping these chemicals into the atmosphere though. They could lead to chronic diseases among humans, or chemically interact with the ozone layer and cause it to deteriorate. Sulfur dioxide in the atmosphere leads to acid rain. Once we start dumping the aerosols in the atmosphere, we will have to keep doing it to sustain the temperature control mechanism. Eventually, the aerosols are going to land back on the Earth. If that happens, it might result in a rapid rise in temperatures. There is also the